

MUHAMMAD RIFQI FATHURRAHMAN

INFORMATION TECHNOLOGY STUDENT | SCIENTIFIC RESEARCH

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Institut Teknologi Sepuluh Nopember (ITS), Surabaya

PROFESSIONAL SUMMARY

Information Technology student with experience in scientific research, scientific writing, technology development, system design, and project coordination. Experienced in conducting research through literature review, problem identification, experimental design, data collection, system testing, data analysis, and scientific report writing. Participated in applied research projects involving microcontrollers, sensors, IoT, renewable energy, automation, and environmental monitoring, including research presented in the Olimpiade Penelitian Siswa Indonesia (OPSI). Also experienced in organizational and event activities, with strong communication, teamwork, leadership, problem-solving, time management, and public-speaking skills. Strong interest in Technology Research & Development, Information Technology Infrastructure, Data Center, and technology-based problem solving.

EDUCATION

INSTITUT TEKNOLOGI SEPULUH NOPEMBER (ITS), Surabaya

Bachelor of Information Technology | 2025 Present

RESEARCH & SCIENTIFIC WRITING EXPERIENCE

OLIMPIADE PENELITIAN SISWA INDONESIA (OPSI) 2024

Researcher & Scientific Writer Applied Physics & Engineering

Project: Rancang Bangun Alat Deteksi Pergerakan Tanah Berkapur Menggunakan Metode Eksperimen Guna Monitoring Performasi Perangkat Mitigasi Bencana

- Conducted scientific research on land movement detection and disaster mitigation in limestone areas.
- Performed literature reviews to establish theoretical foundations and identify relevant previous research.
- Conducted field observation and soil sample collection to obtain supporting research data.

- Developed an experimental research methodology covering hardware and software design, simulation, calibration, testing, and evaluation.
- Collected and analyzed data from MPU6050, soil moisture, and rain gauge sensors to evaluate system performance.
- Conducted sensor calibration, accuracy testing, sensitivity evaluation, error calculation, and system testing.
- Contributed to the preparation of the scientific paper, research documentation, data analysis, and final research report.
- *Bronze Finalist OPSI 2024, Applied Physics & Engineering.*

OLIMPIADE PENELITIAN SISWA INDONESIA (OPSI) 2023

Researcher & Scientific Writer Applied Physics & Engineering

Project: Hybrid Rotary Cylinder Dryer Agricultural Post-Harvest Processing

- Conducted applied research on the development of a hybrid Rotary Cylinder Dryer for agricultural post-harvest processing.
- Conducted literature review, observation, data collection, and experimental research as part of the research process.
- Participated in the formulation of the research problem, objectives, methodology, experimental design, and data analysis.
- Participated in the design, assembly, testing, and evaluation of the research prototype.
- Analyzed experimental results related to moisture content, drying rate, temperature, drying efficiency, and grain quality.
- Utilized Microsoft Excel to support calculations and data analysis.
- Contributed to the preparation of the scientific paper, research report, technical documentation, and presentation materials.
- *Bronze Medal OPSI 2023, Applied Physics & Engineering.*

RESEARCH & SCIENTIFIC WRITING SKILLS

- Scientific Writing
 - Research Methodology
 - Literature Review
 - Problem Identification
 - Research Question & Objective Formulation
 - Experimental Design
 - Data Collection
 - Data Analysis
 - Scientific Report Writing
 - Technical Documentation
 - Research Presentation
 - System Testing & Evaluation
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TECHNICAL SKILLS

Programming

- C, C++

Systems & Development Tools

- Linux, Visual Studio Code, Microsoft Word, Microsoft Excel, Microsoft PowerPoint

Embedded Systems & Electronics

- Arduino, ESP8266, Microcontroller Systems, Sensor Integration, IoT Monitoring Systems, System Design, System Testing, Sensor Calibration

Technical Research

- Technology Implementation, System Design & Prototyping, Experimental Testing, Data Analysis, Technical Writing, Problem Solving

Professional Skills

- Teamwork, Communication, Leadership, Project Coordination, Time Management, Public Speaking, Adaptability

ORGANIZATIONAL & PROJECT EXPERIENCE

UKM PENALARAN ITS

- Staff, Department of Cadre Development | 2026-2027
- Staff Event, Dies Natalis UKM Penalaran | 2026
- Staff Event, Welcome Party Member UKM Penalaran | 2026
- Staff Operational, Visit 1 × Internalisasi | 2026
- Staff Operational, Diklat Member UKM Penalaran | 2026

ELITE X PESRAF FTEIC ITS

- Staff Secretarial | 2026

INCLINATION FTEIC ITS

- Staff Event | 2026
- Supported the planning and execution of event activities.
- Coordinated with team members to ensure activities followed the planned schedule.
- Assisted in managing operational requirements during event implementation.

OKKBK TEKNOLOGI INFORMASI ITS

- Staff Event | 2026

- Supported the preparation and implementation of orientation activities for Information Technology students.
- Coordinated with team members in preparing event activities and operational requirements.
- Assisted in ensuring smooth execution according to the established rundown.

INI LHO ITS

- Staff Event, Submits IMJ | 2026

W-132 & W-133 IT ITS

- Event Division | 2026

CAMPUS EXPO MAN 1 JOMBANG

- Head of Event Division | 2026

DINAMIT ITS

- Event Staff | 2025

ARA 7.0 DEPARTMENT OF INFORMATION TECHNOLOGY ITS

- Event Staff | 2025

MAN 1 JOMBANG

- Coordinator of HUT MAN 1 Jombang Event | 2024
- Head of Jombang Madrasah Research Camp (JMRC) | 2024
- Head of Scientific Youth Organization | 2023-2024
- Treasurer 1 of OSIS | 2023-2024
- Head of P5RA | 2023
- Vice Head of HUT MAN 1 Jombang | 2023
- Public Relations Staff of OSIS | 2022-2023

ACHIEVEMENTS

- 2nd Place DINACOM 11.0 2026, Dian Nuswantoro Computer Club (DNCC), UDINUS.
- Bronze Medal Olimpiade Penelitian Siswa Indonesia (OPSI) 2023, Applied Physics & Engineering.
- Bronze Finalist Olimpiade Penelitian Siswa Indonesia (OPSI) 2024, Applied Physics & Engineering.